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Question Paper ID: 695783

Course Code: CS363IA

USN

12423CB083

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026
(Common to CS and IS)

Compiler Design

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Compare the roles of an assembler and a linker in a language processing system.	02	1	2
1.2	Write the need for input buffering in lexical analysis.	02	2	2
1.3	Do the left-factoring for the following grammar: $S \rightarrow iCtS \mid iCtSeS \mid a$, $C \rightarrow b$	02	2	4
1.4	Explain kernel and Non kernel items with example.	02	1	1
1.5	Consider the Grammar with Productions ($A \rightarrow BC$, $B \rightarrow \text{digit}$ $C \rightarrow \text{digit } C$, $C \rightarrow \epsilon$) with the following SDD, Find the value of A.a for the input 1 2 3 and also find the type of the attribute c and d			
	$A \rightarrow BC$ $C.c = B.b$			
	$B \rightarrow \text{digit}$ $A.a = C.d$			
	$C \rightarrow \text{digit } C_1$ $B.b = \text{digit.lexval}$			
	$C \rightarrow \epsilon$ $C_1.c = \text{digit.lexval} + 1 + C.c$			
1.6	Find the value of A.a for the input 2 4 5 Differentiate between stack and heap memory.	02	1	1

1.7	The least number of temporary variables required to create a three-address code in static single assignment form for the expression $q+r/3+s-t*5+u*v/w$ is _____	02	3	2
1.8	What is the purpose of an activation record in implementing procedure calls? List any two fields it may contain.	02	4	2
1.9	Memory access costs 2 units, register operations cost 1 unit. Find the cost of evaluating $(a+b)*(c+d)$.	02	4	3
1.10	What do the nodes and edges in a flow graph represent?	02	3	2

Part B

Question No	Question	M	CO	BT
2a	A software developer writes a program in C and executes it on a computer. Explain the complete language processing system involved in converting the source program into an executable program, clearly showing the role of the compiler, assembler, linker, and loader.	08	1	2
2b	Compilers must process source programs containing millions of characters efficiently. Explain the need for input buffering. Compare single-buffer and two-buffer schemes with respect to working, efficiency, and implementation complexity.	08	2	2
3a	Consider the following augmented grammar: $S' \rightarrow S$ $S \rightarrow CC$ $C \rightarrow cC$ $C \rightarrow d$ i. Obtain LR(1) items. ii. Construct LALR parsing table. iii. Show the Parsing String "ccdd"	10	4	1
3b	Obtain the LR parsing Table for the following ambiguous grammar $E \rightarrow E+E \mid E-E \mid E*E \mid (E) \mid id$	06	3	4

OR

4a	Given the following grammar: $E \rightarrow 5+T \mid 3-T$ $T \rightarrow V \mid V*V \mid V+V$ $V \rightarrow a \mid b$ i. Is the grammar suitable for predictive Parser? ii. What is the use of left-factoring? Do the left factoring for the above grammar iii. Compute FIRST and FOLLOW sets for each non-terminal iv. Without constructing the parsing table, check whether the grammar is LL(1) or not.	06	3	4
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- v. By constructing the parsing table, check whether the grammar is LL(1) or not.

Consider the following grammar:

$S \rightarrow AA$
 $A \rightarrow Aa \mid b$

4b	i) Construct sets of LR (1) items ii) Construct Canonical LR (1) parsing table iii) Show the parsing steps for the string "baaba"	10	4	4
5a	Consider the grammar generating binary numbers. Design an S-attributed SDD to compute S.val, the 1's complement of the given input string. For example, the 1's complement of a string 1010 should be the binary number 0101(or 101). Construct an annotated parse tree to translate a string 10110 to its complement. $S \rightarrow L$ $L \rightarrow LB \mid B$ $B \rightarrow 0 \mid 1$	08	3	3
5b	Describe the first-fit, best-fit, and worst-fit strategies used in heap allocation. Compare their performance and memory utilization	08	2	1

OR

6a	Consider the grammar for arithmetic expressions $E \rightarrow E + T \mid E - T \mid T$ $T \rightarrow T * F \mid T / F \mid F$ $F \rightarrow (E) \mid \text{digit}$	08	3	5
6b	Design an S-attributed SDD to compute E.count, the total number of operators in the expression. Construct an annotated parse tree for counting the number of operators in the expression $(9+4)*5+1$ Explain stack allocation of storage and illustrate the general structure of an activation record, the information stored in each field	08	3	2
7a	Write quadruples and triples for the expression: i) $a = b[i] + c[j]$ ii) $x = *p + \&y$	08	3	3
7b	Using the backpatching technique, generate intermediate code for the Boolean expression: $(a < b) \text{ OR } (c < d \text{ AND } e < f)$ Show the contents of truelists and falselists at each stage.	08	3	4

OR

8a	Generate three address code for the expression. $a = b[i][j] + c[d[j] + k]$	08	3	3
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Consider the following declarations:

type A = array[1..10] of integer;

type B = array[1..10] of integer;

8b type C = A;

08 4 4

Determine whether the pairs (A,B), (A,C), and (B,C) are equivalent under name equivalence and structural equivalence. Justify your answers.

Apply the following optimizations:

- Constant Folding
- Constant Propagation
- Dead Code Elimination

to the given code.

1. a = 10

2. b = 20

3. c = a + b

4. d = c * 2

5. e = d

6. f = 5 + 10

7. g = f * 2

8. h = g

9a

10 4 4

9b

A target machine has only four registers. Discuss the challenges faced by the code generator and suggest techniques to overcome them.

06 4 5

OR

Given the basic block:

t1 = a + b

t2 = a + b

t3 = t1 * c

t4 = t2 * c

t5 = t3 + t4

x = t5

10a

10 3 3

Construct the DAG and show how it removes common subexpressions. Generate the optimized three-address code.

10b

Illustrate the effect of different instruction formats on code generation and optimization.

06 4 2